

JIAQI WANG (王家琪)

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EDUCATION

The Chinese University of Hong Kong

Sep. 2023 – Sep. 2027

Ph.D. student, Department of Computer Science and Engineering.

Advisor: Prof. James Cheng.

Research Interests: coding agent, reinforcement learning, vision-language models, machine learning theory.

Harbin Institute of Technology (C9)

Sep. 2019 – Sep. 2023

B.Eng. in Artificial Intelligence, Honors College.

INTERNSHIPS

MiniMax, Beijing

Jun. 2026 – Present

Research Topic: Post-Training for Code Expert Models and On-Policy Distillation

- Lead reinforcement learning for long-horizon Code Expert capabilities, including data construction, training, and evaluation pipelines.
- Contribute to on-policy distillation research and final model delivery, supporting the release of the M3.1 and M3 Pro model series.

Tongyi Lab, Alibaba Group, Hangzhou

Oct. 2025 – Apr. 2026

Research Topic: Reinforcement learning for autonomous agents (tool use, memory, and long-horizon reasoning)

- As a core contributor to the RL training framework Trinity-RFT (500+ GitHub stars), implemented scalable training, reward modeling, and evaluation modules.
- Developed and optimized RL algorithms for tool-augmented and memory-augmented agents, improving training stability and sample efficiency. (Paper 1)

FIRST-AUTHOR PUBLICATIONS

1. **Jiaqi Wang**, Wenhao Zhang, Weijie Shi, Yaliang Li, James Cheng “TCOD: Exploring Temporal Curriculum in On-Policy Distillation for Multi-turn Autonomous Agents” *COLM’26*.

TL;DR: Proposes TCOD, a “temporal curriculum” on-policy distillation method that gradually increases trajectory depth to mitigate trajectory-level KL instability in multi-turn agent training, significantly improving stability and performance on multi-step tasks.

2. **Jiaqi Wang**, Kevin Qinghong Lin, James Cheng, Mike Zheng Shou. “Think or Not? Selective Reasoning via Reinforcement Learning for Vision-Language Models” *NeurIPS’25, ICML’25* EXAIT.

TL;DR: Teaches LLM/VLMs “to think or not to think” (TON). We propose a two-stage training strategy combining supervised fine-tuning (with Thought Dropout) and GRPO to reduce unnecessary reasoning steps without performance loss, applied to GUI navigation, 3D object counting, and math problems.

3. **Jiaqi Wang**, Xinyi Wu, James Cheng, Yifei Wang. “A Signed Graph Approach to Understanding and Mitigating Oversmoothing in GNNs” *NeurIPS’25*.

TL;DR: We propose Structural Balance Propagation (SBP), which leverages signed graphs and structural balance theory to prevent oversmoothing, improving classification accuracy on multiple benchmarks.

4. **Jiaqi Wang**, Yuhang Zhou, Zhixiong Zhang, Qiguang Chen, Yongqiang Chen, James Cheng. “DivIL: Unveiling and Addressing Over-Invariance for Out-of-Distribution Generalization” *TMLR’25*.

TL;DR: We define and address the “over-invariance” problem in invariant learning, proposing DivIL by integrating unsupervised contrastive learning with random masking to enhance the diversity of invariant features, improving OOD generalization across NLP, CV, and graph tasks.

5. **Jiaqi Wang**, et al. “VideoASMR-Bench: Can AI-Generated ASMR Videos Fool VLMs and Humans?” *CVPR 2026 Workshop VGBE*.

TL;DR: We propose a video authenticity benchmark suite built on ASMR content to test perceived authenticity under strict audio-visual coupling conditions.

6. **Jiaqi Wang**, et al. “On the Measurement and Efficient Mitigation of Length Generalization Gaps in Large Language Models”, submitted to *NeurIPS’26*.

TL;DR: We propose VCL, theoretically analyze the upper bound of length generalization in transformers, and introduce Virtual Context Learning, which improves length generalization while reducing computational cost.

CO-AUTHORED PUBLICATIONS

1. Dongchi Huang, **Jiaqi Wang**, et al. “PIGDreamer: Privileged Information Guided World Models for Safe Partially Observable Reinforcement Learning” *ICML’25*.

TL;DR: We propose PIGDreamer, a model-based safe RL method that exploits privileged information during training to improve safety and performance under partial observability, extending the CPOMDPs framework with an asymmetric actor-critic architecture.

2. Jing Xu, Daxin Tan, **Jiaqi Wang**, Xiao Chen. “Enhancing Multilingual Speech Generation and Recognition Abilities in LLMs with Constructed Code-switched Data” *ASRU’25*.

TL;DR: We propose CS-LLM, enhancing text-to-speech synthesis in LLMs using only monolingual data.

3. Jing Xu, **Jiaqi Wang**, Daxin Tan, Xiao Chen. “PROST-LLM: PROGRESSIVELY ENHANCING THE SPEECH-TO-SPEECH TRANSLATION CAPABILITY IN LLMs” *ICASSP’26*.

TL;DR: We propose PROST-LLM (Progressive Speech-to-Speech Translation), which progressively enhances speech-to-speech translation capabilities of LLMs.

4. Qiguang Chen, Libo Qin, **Jiaqi Wang**, et al. “Unlocking the capabilities of thought: A reasoning boundary framework to quantify and optimize chain-of-thought” *NeurIPS’24 (Oral)*.

TL;DR: We propose the Reasoning Boundary Framework (RBF), which provides quantitative metrics and guidance for chain-of-thought optimization by defining the Reasoning Boundary (RB) and its composition laws.

5. Binghui Xie, Yongqiang Chen, **Jiaqi Wang**, et al. “Enhancing evolving domain generalization through dynamic latent representations” *AAAI’24 (Oral)*.

TL;DR: We propose MISTS, a mutual-information-based sequential autoencoder that disentangles dynamic and invariant features for Evolving Domain Generalization, achieving better generalization to unseen evolving domains.

6. Zhiyou Xiao, Geng Chen, **Jiaqi Wang**, et al. “NonReasoner: Improving Non-Reasoning in LLM via Test-time Reinforcement Learning”, submitted to *NeurIPS’26*.

TL;DR: We propose the first test-time, label-free reinforcement learning algorithm, finding an efficiency trade-off between thinking and non-thinking modes.

7. Zihui Cheng, Qiguang Chen, Xiao Xu, **Jiaqi Wang**, et al. “Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought” *NeurIPS’25*.

TL;DR: Incorporating visual thoughts, expressing image information into reasoning, improves multimodal chain-of-thought, serving as an intermediate for deeper visual information propagation in large vision-language models.

RESEARCH EXPERIENCE

CSAIL, Massachusetts Institute of Technology (MIT)

Sep. 2023 – Jan. 2025

Research Focus: machine learning theory, graph neural networks, attention mechanisms.

Advisor: Dr. Yifei Wang. Also collaborated with MIT IDSS team, mentored by Dr. Xinyi Wu.

- Researched building trustworthy graph representation learning methods with theoretical guarantees, covering over-smoothing, OOD generalization, and interpretability. (Papers 3, 4)
- Explored combining transformers with graph modeling under various positional encodings (including Alibi, NoPE, and RoPE), and analyzed length generalization upper bounds using Lipschitz constants. (Paper 7)

Torr Vision Group, University of Oxford

Mar. 2025 – Oct. 2025

Research Focus: multimodal agents, reinforcement learning.

Collaborator: Dr. Kevin Qinghong Lin.

- Researched building efficient agents with reinforcement learning, with a focus on GUI agents and video domains. (Papers 2, 5)

COMMUNITY CONTRIBUTIONS

GitHub and MLNLP Community Contributor

Organizer and contributor of three open-source NLP repos, **10.4k+** GitHub stars in total: Paper-Writing-Tips (**4.1k**), Top-AI-Conferences-Paper-with-Code (**2.6k**), and DeepLearning-MuLi-Notes (**3.7k**).